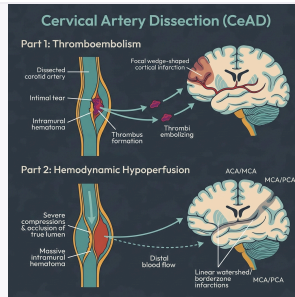
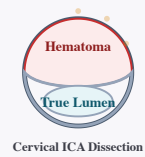
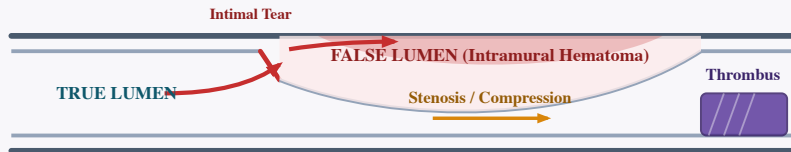


Cervical Artery Dissection



1. Clinical Presentation & Pathophysiology

Ipsilateral Pain & Onset

- **Carotid (ICA):** Frontotemporal/retro-orbital/facial pain (jaw angle).
- **Vertebral (VA):** Severe occipital or posterior neck pain.
- **Onset:** Precedes stroke/TIA by hours to days (median 4 days).

Anhidrosis-Sparing Horner's

- **Signs:** Ptosis/miosis (28–58% of ICA) **without** anhidrosis.
- **Mechanism:** Sweat fibers follow ECA plexus; pupil/eyelid fibers follow ICA.

Neurological Deficits

- **CN Palsies:** CN IX–XII palsies (8–16%) from local ICA compression.
- **VA Territory:** Wallenberg syndrome, cerebellar ataxia, PICA/AICA strokes.

2. Diagnostic Workup

CTA Head/Neck: Shows string sign, dissection flap, pseudoaneurysm, or occlusion.

MRI Neck (T1 Fat-Sat): Crescentic mural signal can support intramural hematoma; timing, artifacts and mimics affect interpretation.

DSA: Reserve for diagnostic doubt or stenting.

Screening: Assess for FMD/connective tissue disease, especially if spontaneous/recurrent.

3. Medical Management: Extracranial vs. Intracranial Dissection

Extracranial Dissection

- **Antithrombotics:** Continue antiplatelet or anticoagulant therapy for at least 3 months (2026 AHA/ASA AIS guideline, Class 2a); the 2024 AHA scientific statement suggests at least 3–6 months, individualized.
- **Antithrombotic choice:** individualize antiplatelet versus anticoagulant therapy for extracranial dissection. ESO expert consensus considers a few weeks of DAPT for selected TIA/minor stroke; this is not a universal 21–90-day regimen, and DOAC evidence is less direct than the randomized VKA comparisons.
- **STOP-CAD:** In occlusive dissection, observational data associate anticoagulation with lower ischemic stroke risk (aHR 0.40); if anticoagulation is chosen, switching to antiplatelet therapy before 180 days seems reasonable to limit major bleeding (not tested in an RCT).
- **IV Thrombolysis:** Reasonable within 4.5 hours in otherwise-eligible patients (Class 2a, LOE C-LD).

Intracranial & Pseudoaneurysms

- **SAH:** Lack external elastic lamina & thin adventitia; rupture risk.
 - **Anticoagulation:** Avoid anticoagulation when SAH is present; intracranial dissection with SAH warrants early endovascular or surgical intervention (ESO 2021). For ischemic presentations without SAH, single antiplatelet therapy is commonly preferred.
 - **IVT Caution:** IVT is reasonable in otherwise-eligible extracranial CeAD, but the risks and benefits of IVT in cases with intracranial extension are not well established (AHA 2024 scientific statement, ungraded).
 - **Intervention:** seek multidisciplinary assessment for exceptional recurrent ischemia despite medical treatment or other complications. Stenosis severity alone is not a routine intracranial-dissection stenting indication; SAH requires a distinct urgent pathway.
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- **Recurrence & Activity:** Long-term CeAD recurrence is low (~1%/yr). Avoid high-risk neck activities (chiropractic neck manipulation, rollercoasters, extreme hyperextension/rotation) for secondary prevention.

4. Landmark Trial & Cohort Evidence

Study / Year	Population & Design	Interventions Compared	Key Outcomes & Clinical Nuance
CADISS 2015	N = 250. Extracranial CeAD. RCT.	Antiplatelet vs. Anticoagulant for 3 months.	• Primary Composite (Ipsilateral stroke or death at 3m): 2.0% vs. 1.0% (p = 0.63). Established clinical equipoise.
TREAT-CAD 2021	N = 194 (PP = 173). Extracranial. RCT.	Aspirin 300mg daily vs. VKA for 3 months.	• Primary Composite (stroke, major hemorrhage, or death to 90d; new ischemic/hemorrhagic MRI lesion at 14d): 23% vs. 15% (Non-inferiority NOT met). Ischemic stroke: 8.0% vs. 0%.
Kaufmann IPD 2024	N = 444. IPD meta-analysis of CADISS + TREAT-CAD.	Antiplatelet vs. Anticoagulant x 90d.	• Primary Composite (Ischemic stroke, major bleeding, or death at 90d): 1.4% anticoagulation vs. 4.4% antiplatelet; OR 0.33 (95% CI 0.08-1.05), P = .06 (not significant). • Ischemic Stroke alone: Significant reduction with anticoagulation (0.5% vs. 4.0%; OR 0.14, p = 0.01), with a non-significant increase in major bleeding (0.9% vs. 0%).
STOP-CAD 2024	N = 3,636. Multicenter observational cohort registry.	Antiplatelet vs. Anticoagulation.	• Temporal Risk: 87% of recurrent strokes occurred in the first 30 days. • Occlusion Benefit: In an interaction analysis, occlusive dissection was associated with lower ischemic stroke risk on anticoagulation (adjusted HR 0.40, 95% CI 0.18–0.88); overall, anticoagulation was associated with a nonsignificantly lower stroke risk by day 30 (adjusted HR 0.71, 95% CI 0.45–1.12). • Transition Strategy: Authors' interpretation: if anticoagulation is chosen, switching to antiplatelet therapy before 180 days seems reasonable, because anticoagulation was associated with more major hemorrhage by day 180 (adjusted HR 5.56, 95% CI 1.53–20.13) but not by day 30 (adjusted HR 1.39, 95% CI 0.35–5.45). A fixed day-30 switch was not tested.

CADISS: *Lancet Neurol.* 2015;14(4):361-7. **PMID: 25684164** (<https://pubmed.ncbi.nlm.nih.gov/25684164/>) | **TREAT-CAD:** *Lancet Neurol.* 2021;20(5):341-350. **PMID: 33765420** (<https://pubmed.ncbi.nlm.nih.gov/33765420/>)

Kaufmann IPD: *JAMA Neurol.* 2024;81(6):630-637. **PMID: 38739383** (<https://pubmed.ncbi.nlm.nih.gov/38739383/>) | **STOP-CAD:** *Stroke.* 2024;55(4):908-918. **PMID: 38335240** (<https://pubmed.ncbi.nlm.nih.gov/38335240/>) | **AHA/ASA:** *Stroke.* 2021;52:e364-e467. **PMID: 34024117** (<https://pubmed.ncbi.nlm.nih.gov/34024117/>) | **AHA Statement 2024:** *Stroke.* 2024;55(3):e91-e106. **PMID: 38299330** (<https://pubmed.ncbi.nlm.nih.gov/38299330/>) | **ESO Guideline 2021:** *Eur Stroke J.* 2021;6(3):XXXIX-LXXXVIII. **PMID: 34746432** (<https://pubmed.ncbi.nlm.nih.gov/34746432/>)